

MARTIN JARAIZ MALDONADO

Professor Emeritus

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EDUCATION

- **Ph.D. in Electronics** (1981), University of Valladolid
Thesis: "Una extensión del modelo de Ebers-Moll para el transistor bipolar en condiciones dinámicas"

RESEARCH EXPERTISE

Computational Modeling Across Disciplines:

- Agent-Based Macroeconomic Modeling (current focus)
- Atomistic Semiconductor Process Simulation (KMC methods)
- Quantum chemistry-based Microkinetic Modeling for Chemical Synthesis

MAJOR CONTRIBUTIONS

Recent Work in Economics (2023-2025)

- **DEPLOYERS Framework:** Developed novel agent-based modeling tool for multi-country economic simulation using FIGARO Input-Output tables
- **Darwinian ABM Methodology:** Pioneered evolutionary approach achieving 100x computational efficiency over traditional DSGE models
- Demonstrated economic forecasting on standard laptops versus supercomputing clusters
- Publications and presentations at IIOA Conferences (Santiago 2024, Malé 2025)

Established Work in Computational Science

- **DADOS Simulator:** Created atomistic KMC process simulator, commercialized globally by Synopsys Inc. and used by more than 30 semiconductor companies including Samsung, Nvidia, IBM, Toshiba, etc.
- **Publications:** 117+ papers, 4,344+ citations (Google Scholar)
- **Industrial Collaborations:** AT&T Bell Labs, Synopsys Inc., NEC Corporation

SELECTED RECENT PUBLICATIONS

Economics & ABM:

- "A novel approach to Economic Forecast on laptops using an Agent-Based Modelling Framework calibrated with multi-country FIGARO I-O tables" (Stuttgart, 2025)
- "DEPLOYERS: An agent-based modeling tool for multi-country real-world data" - IIOA Conference (2024)

Computational Chemistry:

- "DFT-Based Microkinetic Simulations: A Bridge Between Experiment and Theory in Synthetic Chemistry" - Springer (2020)
- ACS Catalysis (2019): Efficient microkinetic modeling protocol achieving >10-fold reduction in computation time

TECHNICAL SKILLS

- **Programming:** C++ (20k+ lines for DEPLOYERS), computational optimization
- **Modeling:** Agent-based systems, Atomistic Kinetic Monte Carlo, Quantum Chemistry calculations

CURRENT RESEARCH

Development of computationally efficient agent-based models for macroeconomic policy analysis, focusing on emergent economic patterns from simple behavioral rules constrained by Input-Output structures. Active on arXiv.org with preprints on evolutionary economic modeling approaches.